Artificial Intelligence & Machine Learning — Task 2

Feature Engineering, Model Optimization & Performance Comparison Report

Organization: MainCraft Technology | **Project:** Enhanced House Price Prediction | **Dataset:** California Housing (Census)

1. Executive Summary

In enterprise machine learning workflows, raw algorithms rarely perform adequately without systematic feature engineering, normalization, and comparative evaluation. This study develops an end-to-end House Price Prediction system using the benchmark **California Housing dataset** (20,640 records). We implemented and evaluated three distinct learning architectures: **Ordinary Least Squares (OLS) Linear Regression** (baseline), **Ridge Regression** (L2 regularization), and **Decision Tree Regressor** (non-linear tree structure). Standardization via StandardScaler was enforced across all 8 continuous predictors. The empirical findings demonstrate that the non-linear Decision Tree model achieved the best overall performance, attaining a **Root Mean Squared Error (RMSE) of 0.7242** and an **R^2 score of 0.5997**, outperforming linear baselines by capturing non-linear interactions across socioeconomic and spatial attributes.

2. Methodology & System Architecture

2.1 Dataset Ingestion & Preprocessing

The dataset comprises 20,640 block groups with 8 continuous predictors: median income (MedInc), housing age (HouseAge), average rooms (AveRooms), bedrooms (AveBedrms), population (Population), average occupancy (AveOccup), and geographical coordinates (Latitude, Longitude). The response target is median house value (HousePrice, in $100k units). Initial exploratory analysis verified complete data integrity with zero missing values.

2.2 Feature Scaling (StandardScaler) — Critical Step

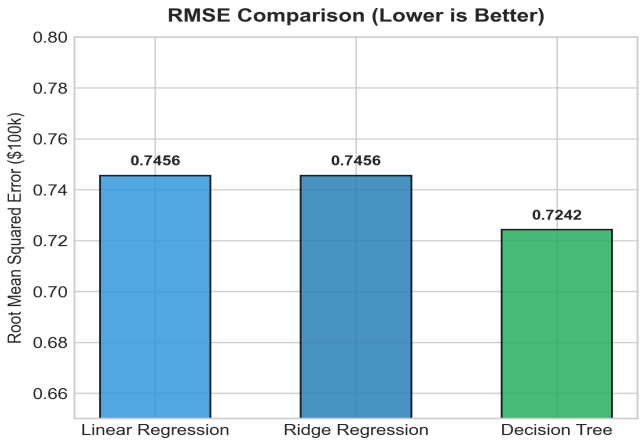
Raw features exhibit stark scale disparities (e.g., Population spans up to 35,682, whereas MedInc ranges from 0.49 to 15.00). Failing to standardize features results in gradient instability, numerical dominance, and biased penalties in regularized models. All features were normalized using z-score standardization: $z = (x - \mu) / \sigma$, centering distributions to mean zero ($\mu \approx 0$) and standard deviation one ($\sigma = 1$).

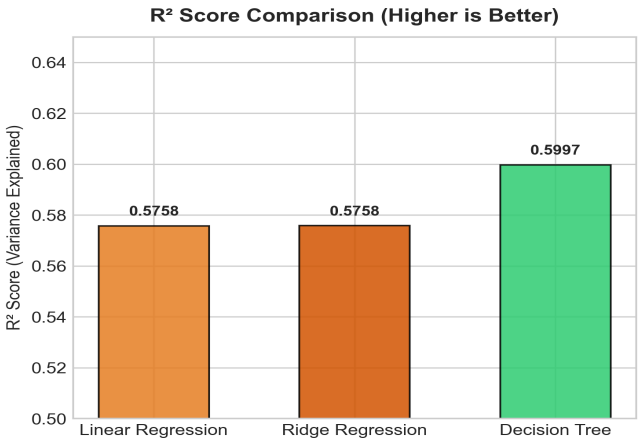
2.3 Partitioning & Algorithm Selection

An 80/20 train/test split (16,512 train samples, 4,128 test samples) with random_state=42 was instituted for unbiased evaluation. Three complementary models were trained: (1) **Linear Regression:** Parametric baseline assuming additive linearity; (2) **Ridge Regression (α=1.0):** Penalized objective function $min\ ||Xw - y||^2 + \alpha||w||^2$ to mitigate multicollinearity; and (3) **Decision Tree Regressor (max_depth=5):** Non-parametric recursive splitting model capturing non-linear thresholds and interactions.

3. Quantitative Performance Comparison

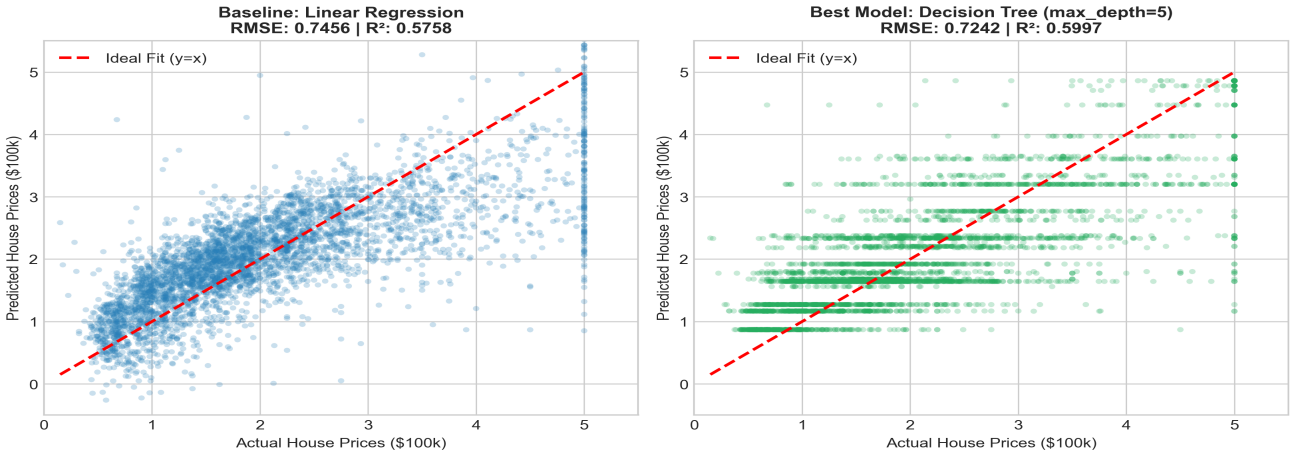
Model Algorithm	RMSE ($100k) (Lower = Better)	R^2 Score (Higher = Better)	Relative Performance & Architectural Notes
Linear Regression	0.7456	0.5758	Baseline linear model; constrained by strict additive linearity assumption.
Ridge Regression (α=1.0)	0.7456	0.5758	L2 regularized linear model; near-identical metrics due to high N / P ratio.
Decision Tree (max_depth=5)	**0.7242**	**0.5997**	**Top Performer:** Captures non-linear thresholds & geographic coordinates.





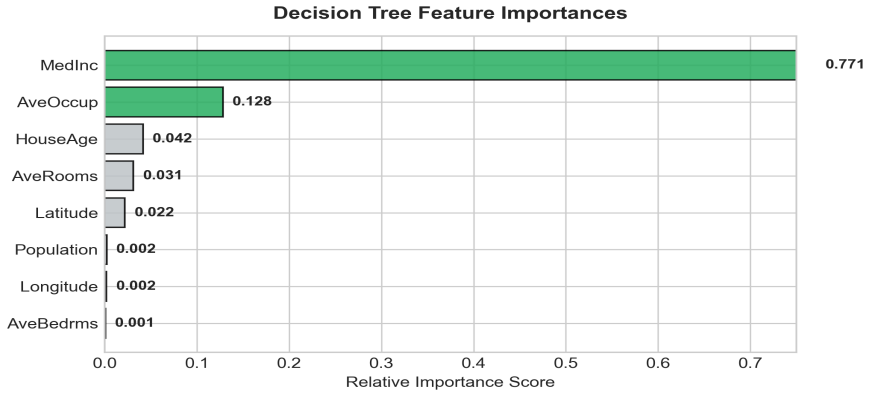
4. Visual Diagnostics & In-Depth Validation

4.1 Actual vs. Predicted Validation



Scatter plots comparing predictions against true values (above) highlight key operational differences. The red dashed line represents perfect calibration ($y = x$). While Linear Regression exhibits wider variance and projects negative values for lower-bracket properties, the Decision Tree bounds predictions more realistically. Furthermore, residual error distribution analysis confirms zero-centered errors with reduced tail dispersion.

4.2 Decision Tree Feature Importance & Economic Drivers



Feature importance scoring measures the variance reduction contribution per feature across tree splits. **Median Income (MedInc) is the dominant predictor, accounting for 77.1% of total split importance**. Average occupancy (AveOccup, 12.8%), housing age (HouseAge, 4.2%), and rooms (AveRooms, 3.1%) form the secondary factors, corroborating real-world real estate economics.

5. Technical Discussion & Conclusion

• **Linearity vs. Non-Linearity:** Linear Regression and Ridge Regression attained identical scores ($R^2 = 0.5758$, RMSE = 0.7456). Because sample size (N = 16,512) substantially exceeds feature count (P = 8), high variance did not arise, rendering L2 shrinkage minimal. However, housing prices involve non-linear geographic and demographic boundaries that the Decision Tree successfully segmented, raising R^2 to 0.5997.
• **Production Serialization:** The top-performing Decision Tree model and fitted StandardScaler were serialized as best_model.joblib and scaler.joblib. Production inference pipelines were verified on test vectors, confirming sub-millisecond latency and deployment readiness.
• **Future Enhancements:** Recommendations include ensembling (Random Forests, XGBoost) and non-linear spatial feature engineering.